

004.02 Data Quality Policy

 Page Status - In Development

Description

This intends to describe the tenets of a data quality policy to be compared and executed against. It stands not as the implementation, but guidelines for implementation as quality is integrated within pipelines, processes, diligence and activity.

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What is a Data Quality Policy?

A data quality policy sets out an approach for managing data as an asset. It intends to ensure that data is:

- Held securely and privately
- Obtained fairly, lawfully
- Recorded accurately and reliably
- Used effectively and ethically
- Shared appropriately and legally

The focus of this page to document a framework for recording data accurately and reliably with later focus on security and privacy.

Why implement a Data Quality Policy?

There are many challenges related to data quality issues, a few are mentioned related to improved data quality:

- Ensure confidence that all data used for decision making is highest obtainable quality
- Allow business to confidently chase business opportunities

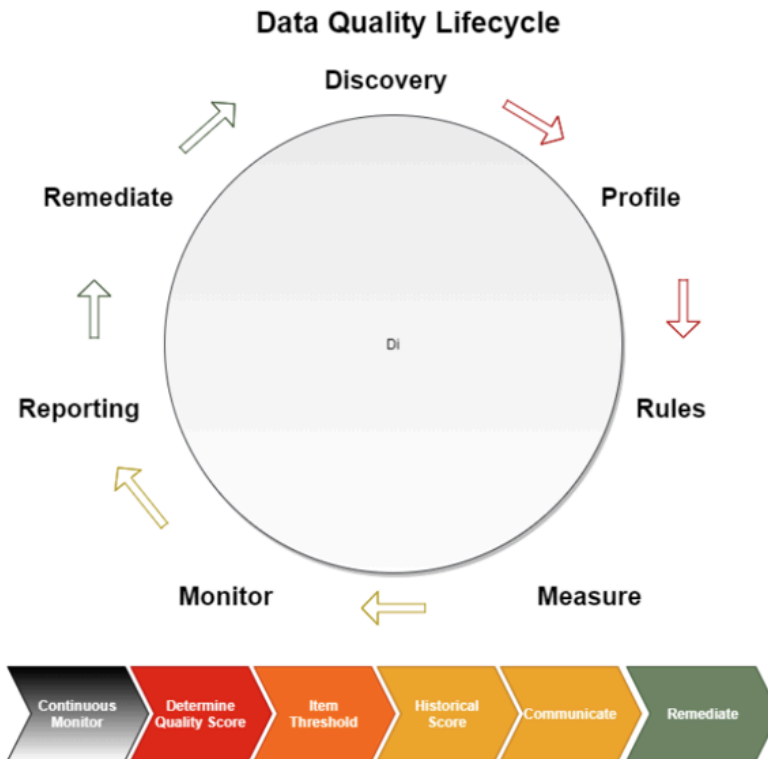
How do we measure our Data Quality Policy?

- Measurement is applied based on a grading scale:
 - F No proactive improvements
 - D Limited data analysis and correlation, correction and cleansing
 - C Start to address root cause defects with quality training
 - B Proactively prevent future data defects, add more disciplines. Metrics in place to measure number of defects
 - A Optimization cycle by continuously monitoring, communicating and improving a data defect

Data Quality Life Cycle

The process of implementing data quality can be considered a continuous cycle, either as an initial test or continuous measurement through a data quality life cycle.

The monitoring process itself is also a continuous recycle with a historical and current comparison activity.



Discovery

Discovery aims to determine all metadata concerning the source, analysis of procurement/ingress process, is the data relevant/usable and can it be received in a stable, monitored process

Source Consumption Analysis

- How is the data received? Is it a reliable pattern?
- Is the data relevant? Should it be used?

Attribute Analysis

- Duplicate Count. Do we have attributes that are the same?
- Documented vs Inferred Data Types
- Precision and Scale of Numerical values
- Profile columns with LOV (List of Values)
- Columns that can be Null
- Column(s) that should be unique

Functional Dependency

- Determine extent of which values in one column (dependent) depend on values in another column (determinant)
- Join Analysis and Relationship Analysis

- Determine extent of commonality between multiple attributes belonging to different entities.
- Identify common records as well as orphan records in terms of count and percentage.

Data Domain Discovery

- Predefined or user defined object based on the semantics of the column or a column name.
- Use the discovery to protect fields (like SSN for privacy or Comments to remove from moving or Notes)
- Rule Based Advanced Profiling
- Predefined business rules are identified and applied as an exercise.

Profiling

Profiling involves analysis of the data, comparisons of actuals to metadata, calculation of statistics and identifying the essential or necessary measures of quality

Timeliness

Does data represent reality within a reasonable time?

Data should be captured as quickly as possible after event and be available for intended use within time period. Measured by the time between when the information is expected and when its ready for use.

Number of Records Processed

- Total Records
- Time Period to Process Records
- Number of Records Processed/Total Records/ Time Period to Process
- Current Time - Update Time: Difference between Update and
- Time of Storage
- Storage Time - Update Time / Total Time: Time length for which data remains valid

Completeness

- What data is missing or unusable?
- Do we have all of the data we expect?
- Identification of missing, incomplete or invalid records provides an indication of data quality within the capture process
- Mandatory attributes that require a value
- Optional attributes which may have a value based on conditions
- Inapplicable attributes which may not have a value

- Entity completeness - requires that all instances exist for all business entities. - - All records or rows are present
- Relationship completeness - Referential integrity exist
- Attribute completeness - All business attributes for each business entity exist
- Domain completeness - Demand that all business attributes contain allowable values and that null values can be differentiated from missing.
- Number of Null Values
- Number of Null Values for a Required Entry
- Number of White Space Rows
- Number Junk Sequences
- Number Not Populated Records
- Number Actual Total Size / Expected Total Size = Completeness
- Min of Record
- Max of Record
- List of Value (LOV) Identification

Consistency

What data values receive conflicting Information?

- Refers to data values in one set being consistent with values in another set. No matter where you look in the system, there are not contradictions in your data for the same piece of data.
- Consistency refers to data values, logic or calculations being aligned to their standards and definitions regardless of source
- Number of Inconsistent Values / Total Number of Values
- Total Number of Values Violating Constraints/Rules

Uniqueness

- Every business entity must be unique
- Every business entity must have only one unique identifier
- Each business attribute must have only one unique name
- Each business attribute must have only one unique domain and is not overloaded
- De-dup - Define match rules and consolidation criteria for winner/lose
- Uniqueness between entities and attributes.
- Does the same data live elsewhere
- How unique on its own, across entities, compared to other
- Number Unique Records

- Number of Duplicate Records
- Number of Distinct Values
- Distance from other fields-Jaccard
- Distance from entities
- Frequency Distribution
- Entity uniqueness (bray)
- Column uniqueness

Integrity

Measure of the existence, validity, structure, content and other characteristics of data.

- Are e-mail addresses in correct form?
- Number of Orphan Records
- Number of Referential Integrity Errors
- Number Incorrect Structure (Email Address)
- Degree of conformity to defined data relationships (primary, referential)

Accuracy

- Inaccurate dates or product codes or range analysis.

Accuracy is calculated on top of the context of the information for which the data contributes.

For example, an effective date should always precede an expiration date.

- Are the values in the correct data type?
- Total # of Transformations
- % of Total Transformations
- Ratio of Data Errors
- Outside of applied rule set (City = Boston, State = CA

Reliability

- How is the data received?
- Is the pattern identified? Reliable? Scalable?
- Number of Rows Unable to Process

Relevance

- Is the data needed? Useful?
- How much of the data is being used and at what cadence?
- Cost of Storage - Invalid Data
- Amount of Dark Data

Reasonability

- Does the data volume relate to expected? Size/Volume of Incoming to Previous Validity
- Conformance to a set of allowable values or range of values.
- Data is valid if it conforms to syntax (format, type, range, referential integrity) of its definition
- Assess format, pattern, legibility, case, expectations
- Confirming data type (phone #'s are numeric)
- Comparisons to expected incoming
- Comparisons to thresholds (tolerances, limits, % variance to prior)
- Conforming to valid values, not conforming

Data Quality Rules

Identify Key Data Elements (KDE) that are most important to business for each domain. Data quality rules should be developed for each Key Data Element (KDE) for each relevant Data Quality Measure

There are four basic categories for Data Quality Rules:

1. Business Entity Rules
2. Data Validity Rules
3. Data Dependency Rules
4. Attribute Dependency

Business Entity Rules

- Uniqueness
 - Every instance of a business entity has its own unique identifier (primary)
 - Primary can never be null
 - Rule number three applies only to composite or concatenated keys
 - Composite primary key can contain one or more foreignCardinality
 - Number of times business entity can be related to another entity

Data Validity Rules

- List of values, such as US States
- Range of Values (1-100)
- Constraint on Values (<130)
- Set of allowable characters
- A pattern, such as a date

Data Dependency Rules

Data dependency rules apply to data relationships between 2 or more business entities as well as between business attributes. Existence of data relationship depends on the state (condition) of another entity.

For example

- Orders cannot be placed for a customer who is delinquent. Existence of one data relationship mandates that another data relationship also exists.
- A salesperson must be associated with an order
- Existence of one data relationship prohibits the existence of another relationship. An employee assigned to a project cannot be enrolled in a training program.

Attribute Dependency

- Value of one business attribute depends on state of the entity in which attribute exists.
 - Ex: Loan Amount > 0 if loan status = Funded
- Correct value of one attribute depends or is derived from the values of two or other attributes.
 - Ex: Pay Amount must equal hours * rate
- Allowable value of one attribute is constrained by the value of one or more other attributes in the same business entity.
 - Ex: When loan type code = ARM4 and funding date is prior to 10/1, the ceiling interest rate cannot exceed 6 percent
- The existence of one attribute value prohibits the existence of another attribute value in the same business entity.
 - Ex: Monthly Salary > 0, then commission rate must be null

Measure

Measurement requirements implementation of the identified measures at the appropriate and identified dimensions.

Monitor

Monitoring is the process of initial and on-going execution of the rules and measures process.

Reporting

Reporting requires identification and implementation to measure and publicize the results of initial and on-going monitoring.

The adherence to defined rules and measuring process must create an actionable and identifiable task for adjustment with assigned remediation and communication.